

Wizard Code-Tracing Problem:

Answer:

Gandalf has learned Fireball!
Gandalf has learned Frostbolt!
Saruman has learned Fireball!

--- Gandalf begins casting! ---
Cannot cast Lightning Bolt. Spell is unknown.
Casting Frostbolt with power 40!

--- Saruman begins casting! ---
Casting Fireball with power 60!

Total damage dealt by Gandalf: 40
Total damage dealt by Saruman: 60

Final spellbooks: {'Fireball': 50, 'Frostbolt': 40} and {'Fireball': 60}

PlayerProfile Programming Problem:

Answer:

```
class PlayerProfile:
    def __init__(self, player_name, max_health):
        self.player_name = player_name
        self.max_health = max_health
        self._health = max_health # Private attribute for current health
        self._level = 1           # Private attribute for current level

    def __str__(self):
        # Provides a clean, readable output when the object is printed
        return (f"--- Player: {self.player_name} ---\n"
                f"Level: {self._level}\n"
                f"Health: {self._health}/{self.max_health}")

    # --- Traditional Accessor and Mutator for Health ---
    def get_health(self):
        """Accessor: Returns the current health."""
        return self._health

    def set_health(self, new_value):
        """Mutator: Sets the health with validation."""
        if new_value < 0:
            self._health = 0
```

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        elif new_value > self.max_health:
            self._health = self.max_health
        else:
            self._health = new_value

# --- Example Usage ---
player1 = PlayerProfile("Kratos", 150)
print(player1)

current_health = player1.get_health()
player1.set_health(current_health - 30)
print(player1)

player1.level = -10
print(f"Current level after hack attempt: {player1.level}")

# Create a new profile for a different character with lower health
player2 = PlayerProfile("Elara", 80)
print(player2)

player2.level = "five" # The setter should reject this non-integer value

current_health = player2.get_health()
player2.set_health(current_health - 100) # This should be clamped at 0, not
become negative
print(player2)

player2.set_health(40) # Directly setting the health
print(player2)

```

 PlayerProfile
+ player_name: String - level: int -health: int + max_healht: int
+ __init__(name:String, max_health: int) + __str__(): String + get_health(): int +set_health(new_value: String)